

Multi-Angle ISAR Image Fusion for Aerial Target Recognition Using Image Transformer

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Abstract—Inverse Synthetic Aperture Radar (ISAR) imaging has become a key technology for airborne target recognition, but due to the dependence of scattering patterns on the viewing angle, single-angle observation often suffers from insufficient discriminative information. This paper proposes a novel multi-angle fusion framework based on an image Transformer, utilizing multiple observation angles to comprehensively capture target features and improve ISAR target classification accuracy. Based on the designed image Transformer model, we investigate feature-level and decision-level fusion strategies, including feature-level entropy-weighted fusion and splice fusion, as well as decision-level average voting. Experiments were conducted on a dataset containing eight aircraft types and 7200 ISAR images, using different numbers of angular intervals and different fusion methods. The results show that our multi-angle fusion method can achieve a classification accuracy of up to 98.75% when using five observation angles and the entropy-weighted fusion strategy, significantly outperforming the single-angle recognition accuracy of 86.83%. The experimental results verify that multi-angle information fusion can significantly improve the robustness and accuracy of classification, achieving optimal performance with four to five observation angles.

Keywords—ISAR image recognition, Image Transformer, multi-angle fusion, aerial target recognition, deep learning

I. INTRODUCTION

Inverse Synthetic Aperture Radar (ISAR) target recognition has emerged as a critical technology in modern radar systems, offering distinctive advantages in all-weather surveillance and autonomous target identification. G. Dong et al. [1] proposed a new target enhancement method that uses perturbation range profiles to image targets and form new ISAR images. J. Duan et al. [2] developed an automated deep neural network architecture that transforms the identification of anomalies in ISAR images into a sequential classification task. W. Zhao et al. [3] applied transfer learning to ISAR ship imagery classification.

Concurrent with ISAR recognition advances, Vision Transformers (ViTs) have fundamentally challenged CNNs in image classification. Dosovitskiy et al. [4] introduced the seminal ViT architecture treating images as patch sequences

with self-attention mechanisms, achieving competitive performance on image classification benchmarks. The transformer's self-attention naturally captures relationships between scattering points, proving advantageous for ISAR analysis. Recent work has refined transformer architectures for remote sensing applications. H. Yu et al. [5] developed MSTNet, a multilevel spectral-spatial transformer network for hyperspectral image classification. Y. Wang et al. [6] proposed spectral-spatial-temporal transformers for hyperspectral image change detection. H. Yuan et al. [7] proposed an end-to-end recognition framework for maneuvering space targets in the complex value (CV) domain based on a transformer model, which achieved accurate classification of ISAR images. Despite these advances, most existing methods focus on single-view recognition without fully exploiting multi-angle information.

Multi-view fusion has shown substantial benefits across computer vision and remote sensing. M. Zhou et al. [8] presented a general spatial-frequency learning framework for multimodal image fusion, establishing theoretical foundations for effective multi-view integration. W. Li et al. [9] developed an asymmetric feature fusion network for hyperspectral and SAR image classification, addressing modality imbalance challenges. G. Li et al. [10] proposed a novel transformation method based on the SelectionGAN network, which improves target recognition capabilities by fusing cross-viewpoints of optical and radar images. G. Zhao et al. [11] combined CNN and transformers hierarchically for joint hyperspectral and LiDAR classification, validating hybrid architectures' advantages. Y. Wan et al. [12] proposed a multimodal adaptive feature fusion network (MAFF-Net) to accurately classify low, slow, and small (LSS) targets in the low-altitude airspace.

For ISAR specifically, existing methods lack systematic investigation of optimal angle quantities and effective fusion strategies using transformer architectures. To address these challenges, we propose a comprehensive multi-angle ISAR fusion framework based on Image Transformers. The framework includes various fusion strategies and configurations for different numbers of corner domains. Experimental validation demonstrates that our approach achieves 98.75% accuracy, representing a 11.92 percentage point improvement over single-angle baselines.

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II. METHODOLOGY

A. Image Transformer for ISAR Feature Extraction

The feature extraction backbone employs an Image Transformer architecture adapted for ISAR imagery, as shown in Fig. 1.

The Image Transformer network comprises five key components. The input layer receives $224 \times 224 \times 3$ -pixel images with preprocessing, which are divided into 196 non-overlapping 16×16 -pixel patches by the patch embedding layer. Each patch is mapped to a 768-dimensional feature vector via convolutional projection with stride 16. The positional encoding module generates 768-dimensional vectors for each patch position, fused with patch features through element-wise addition. The core Transformer encoder consists of six stacked blocks, each containing multi-head self-attention and feedforward network sublayers with residual connections and layer normalization. The multi-head self-attention employs 12 heads to capture global spatial relationships between scattering points. The feedforward network performs nonlinear transformation with output dimension 768, hidden dimension 3072, and GELU activation.

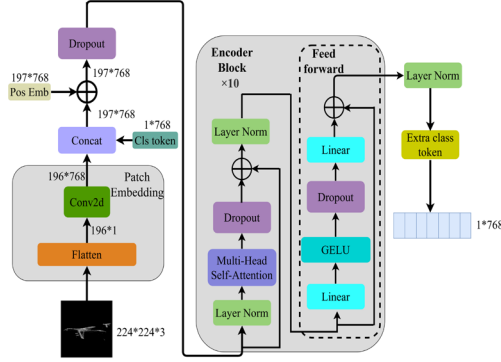


Fig. 1. Image Transformer network structure diagram.

As the core component of Transformer architecture, the multi-head self-attention layer computes weighted representations for sequence elements by evaluating pairwise relationships among all positions. This mechanism employs parallel computation across multiple attention heads, enabling the model to extract features from diverse representation subspaces simultaneously. Its computational process detailed in the following steps:

Step 1 - feature projection: the input matrix, defined as is projected into the Query (Q), Key (K) and Value (V) space using different learnable parameters W_Q , W_K and W_V , which are computed using the following formulas:

$$Q = XW_Q, K = XW_K, V = XW \quad (1)$$

where W_Q , W_K , and W_V denote the query mapping matrix, key mapping matrix, and value mapping matrix, respectively.

Step 2 - attention score computation and weighted aggregation: to quantify the relevance between different sequence positions, we calculate attention scores through the following operation:

$$\text{Attention} = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) \quad (2)$$

where K^T denotes the transpose of the key space and d_k denotes single-head attention dimension. The resulting score captures the correlation strength between temporal positions i and j , where higher values indicate stronger dependencies. The raw attention scores undergo SoftMax normalization to produce probability distributions.

Step 3 - weighted aggregation: the normalized attention weights are applied to the value representations through weighted summation:

$$\text{head}_i = \text{Attention} \cdot V = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (3)$$

where head_i incorporates global contextual information at each temporal position, as the attention mechanism allows each step to selectively focus on relevant parts of the entire sequence.

B. Feature-Level Fusion

As shown in Fig. 2, the feature-level fusion network adopts a distributed extraction, intelligent fusion, and unified decision-making architecture. The system consists of parallel image Transformer feature extractors, a feature projection layer, an intelligent fusion strategy module, and a shared classifier.

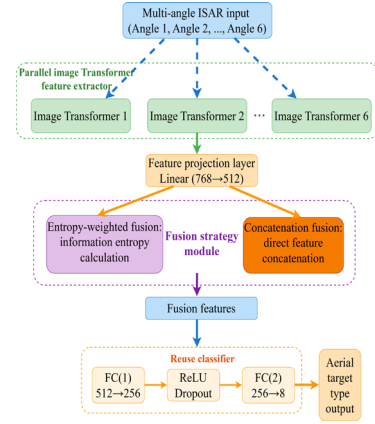


Fig. 2. Feature-level fusion network architecture.

1) Entropy-Weighted Fusion

We propose an adaptive fusion mechanism based on Shannon entropy:

Entropy computation and weight assignment: calculate Shannon information entropy and compute adaptive fusion weights:

$$H^k = - \sum_{i=1}^{512} \tilde{f}_i^k \log(\tilde{f}_i^k + \epsilon), w^k = \frac{1/H^k}{\sum_{j=1}^K 1/H^j} \quad (4)$$

where H^k represents the information entropy of the k th angle's features, $\tilde{f}_i^k$ denotes the i th element of the normalized

feature vector from angle k , and ϵ is a small constant for numerical stability to avoid logarithm of zero. w^k represents the adaptive fusion weight for angle k .

Weighted fusion: combine features through weighted summation:

$$f_{\text{fused}} = \sum_{k=1}^K w^k f^k \quad (5)$$

where f_{fused} denotes the final fused feature vector, f^k is the original feature from angle k .

The designed feature-level fusion network is based on the forward-looking feature fusion design concept, enabling the network to consider the complementarity of viewpoint information during feature learning and to learn stable and invariant feature representations for each type of aerial target.

2) Splice Fusion

We concatenate all angle features and apply dimensionality reduction:

$$f_{\text{concat}} = [f^1; f^2; \dots; f^K], f_{\text{fused}} = f_{\text{concat}} W_{\text{reduce}} + b_{\text{reduce}} \quad (6)$$

where f_{concat} represents the concatenated feature vector from all K angles, W_{reduce} is the learnable weight matrix for dimensionality reduction, b_{reduce} is the bias vector, and f_{fused} is the final fused feature vector after reduction.

C. Decision-Level Fusion

As shown in Fig. 3, the designed decision-level fusion network adopts a completely independent parallel decision and average ensemble architecture. It processes data in parallel through multiple independent image Transformer networks, ultimately achieving intelligent fusion at the decision level.

For decision-level fusion, each angle k maintains an independent image Transformer backbone and classifier:

$$\text{logits}^k = h^k W_2^k + b_2^k, \text{ logits}_{\text{final}} = \frac{1}{K} \sum_{k=1}^K \text{logits}^k \quad (7)$$

where logits^k represents the classification logits from angle k , $\text{logits}_{\text{final}}$ is the averaged logits across all angles.

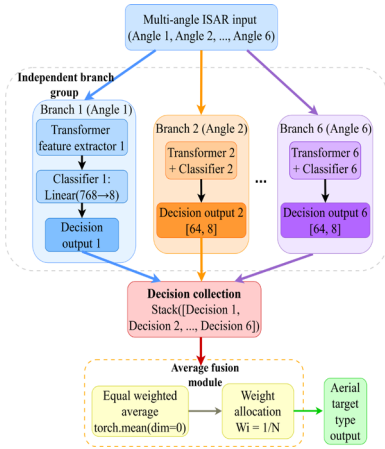


Fig. 3. Decision-level fusion network architecture.

The combined results of multiple independent classifiers with reasonable accuracy are usually superior to those of a single classifier, as averaging reduces prediction variance and improves decision stability.

III. EXPERIMENTAL DESIGN

A. Dataset Description

We constructed a comprehensive ISAR dataset comprising eight aircraft classes: C17, Yun-9, E2, E3, B52, C5, RC-135, and P-3C. The dataset contains 7,200 ISAR images totaling 900 samples per class. Azimuth coverage spans six intervals, the specific angle range correspondence table is shown in Table I.

TABLE I ANGLE RANGE CORRESPONDENCE TABLE

Angle range	0-29°	30-59°	60-89°	90-119°	120-149°	150-179°
Number	0	1	2	3	4	5

Fig. 4 shows the ISAR images of different types of aerial targets at azimuth angles of 0-29° and 150-179°.

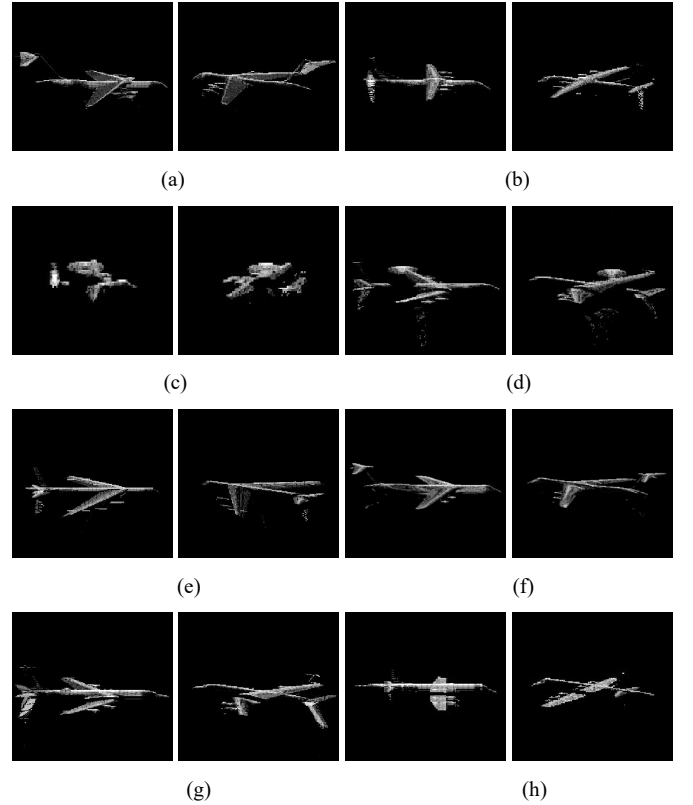


Fig. 4. ISAR images of airborne targets at azimuth angles of 0-29° and 150-179°. (a) C17. (b) Yun-9. (c) E2. (d) E3. (e) B52. (f) C5. (g) RC-135. (h) P-3C.

As can be seen from the example dataset, the overall contours of ISAR images of the same target differ significantly at different angles, posing a considerable challenge for the designed model and the fusion strategy employed to capture angle-invariant features of airborne targets.

B. Implementation Details

The experiment was conducted using the PyTorch framework, with the dataset divided into a 6:2:2 ratio. The model extracted images from specific angular ranges, forming multi-angle image tuples centered around the sampling points. The total number of epochs was set to 150, the batch size to 64, the learning rate to $1e-4$, weight decay to $2e-4$, dropout to 0.2, and the initial random seed to 42. Experiments were conducted using three different fusion methods under varying numbers of angular ranges, resulting in 15 experiments. An additional experiment was performed with a single angular range, bringing the total to 16 experiments. Each experiment was repeated 10 times, and the average results were taken.

IV. TEST RESULT ANALYSIS

The specific configuration of each experiment and the corresponding average test accuracy are shown in Table II.

TABLE II EXPERIMENTAL CONFIGURATION AND AVERAGE ACCURACY.

Experiment ID	Fusion Level	Fusion Method	Fusion Angle Number	Average accuracy
0	Feature level	weighted	[0,1]	93.75%
1	Feature level	splice	[0,1]	87.92%
2	Feature level	weighted	[0,1,2]	95.83%
3	Feature level	splice	[0,1,2]	87.92%
4	Feature level	weighted	[0,1,2,3]	97.92%
5	Feature level	splice	[0,1,2,3]	94.58%
6	Feature level	weighted	[0,1,2,3,4]	98.75%
7	Feature level	splice	[0,1,2,3,4]	97.92%
8	Feature level	weighted	[0,1,2,3,4,5]	97.50%
9	Feature level	splice	[0,1,2,3,4,5]	97.08%
10	Decision level	average	[0,1]	93.33%
11	Decision level	average	[0,1,2]	97.08%
12	Decision level	average	[0,1,2,3]	97.08%
13	Decision level	average	[0,1,2,3,4]	95.00%
14	Decision level	average	[0,1,2,3,4,5]	97.92%
15	-	-	Single angle	86.83%

The overall performance comparison results are shown in Fig. 5. The results indicate that multi-angle fusion significantly improves classification accuracy, outperforming single-angle recognition. The best performance reached 98.75%, an improvement of 11.92 percentage points compared to the single-angle baseline (86.83%). Feature-level weighted fusion consistently outperformed other strategies, validating the effectiveness of the entropy-based adaptive weighting method.

Fig. 6. shows that the average accuracy increases rapidly from 2 to 4 angles (from 91.67% to 96.81%), and then stabilizes at approximately 97.22% between 4 and 5 angles. Beyond 5 angles, the performance does not change significantly, indicating diminishing returns from increasing the number of

observation angles. The optimal number of angles for fusion is between 4 and 5.

Fig. 7. shows the performance curves of different fusion methods. It is clear that the entropy-weighted fusion method achieved the highest peak performance across five angular intervals and demonstrated excellent adaptability through dynamic weight adjustment. The performance improvement of the splice fusion method was relatively slow, while the decision-level fusion method showed stable performance, reaching a peak of 97.92% at six angles.

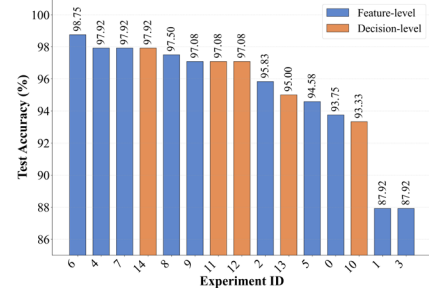


Fig. 5. Overall performance comparison histogram.

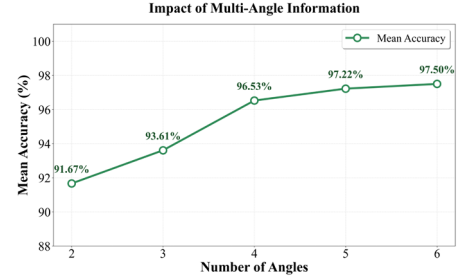


Fig. 6. Line chart of mean accuracy varying with the number of fusion angles.

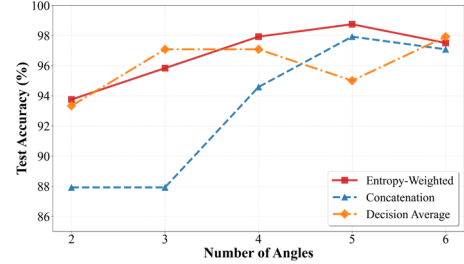


Fig. 7. Fusion method comparison.

V. CONCLUSION

To address the challenge of angle sensitivity in ISAR-based airborne target recognition, this paper proposes a multi-angle fusion framework leveraging an image Transformer. By employing an entropy-weighted fusion mechanism, the framework adaptively integrates complementary angular information, significantly enhancing feature representation. Experimental results on an eight-aircraft ISAR dataset demonstrate that the optimal 5-view configuration achieves a recognition accuracy of 98.75%, marking an 11.92% improvement over single-view methods. Furthermore, the study identifies that 4 to 5 viewing angles provide the best trade-off between recognition performance and resource efficiency, offering a robust reference for multi-view sensor fusion in practical observation scenarios.

REFERENCES

- [1] G. Dong and H. Liu, "Target Recognition in ISAR Image via Range Profile Perturbation Imaging," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1-11, 2024, Art no. 5227811.
- [2] J. Duan, Y. Ma, L. Zhang and P. Xie, "Abnormal Dynamic Recognition of Space Targets From ISAR Image Sequences With SSAE-LSTM Network," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1-16, 2023, Art no. 5102916.
- [3] W. Zhao, A. Heng, L. Rosenberg, S. T. Nguyen, L. Hamey and M. Orgun, "ISAR Ship Classification Using Transfer Learning," 2022 IEEE Radar Conference (RadarConf22), New York City, NY, USA, 2022, pp. 1-6.
- [4] A. Dosovitskiy, L. Beyer, A. Kolesnikov, et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2021.
- [5] H. Yu, Z. Xu, K. Zheng, D. Hong, H. Yang and M. Song, "MSTNet: A Multilevel Spectral-Spatial Transformer Network for Hyperspectral Image Classification," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1-13, 2022, Art no. 5532513.
- [6] Y. Wang et al., "Spectral-Spatial-Temporal Transformers for Hyperspectral Image Change Detection," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1-14, 2022, Art no. 5536814.
- [7] H. Yuan, H. Li, Y. Zhang, C. Wei and R. Gao, "Complex-Valued Multiscale Vision Transformer on Space Target Recognition by ISAR Image Sequence," in *IEEE Geoscience and Remote Sensing Letters*, vol. 21, pp. 1-5, 2024, Art no. 4008305.
- [8] M. Zhou et al., "A General Spatial-Frequency Learning Framework for Multimodal Image Fusion," in *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 7, pp. 5281-5298, July 2025.
- [9] W. Li, Y. Gao, M. Zhang, R. Tao and Q. Du, "Asymmetric Feature Fusion Network for Hyperspectral and SAR Image Classification," in *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 10, pp. 8057-8070, Oct. 2023.
- [10] G. Li, X. Yang, Y. Zhang and H. Ren, "A Novel Cross-View Fusion Approach for Ship Targets Based on ISAR and Optical Images," *IGARSS 2025 - 2025 IEEE International Geoscience and Remote Sensing Symposium*, Brisbane, Australia, 2025, pp. 8764-8768.
- [11] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan and B. Jeon, "Joint Classification of Hyperspectral and LiDAR Data Using a Hierarchical CNN and Transformer," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1-16, 2023, Art no. 5500716.
- [12] Y. Wan, J. Chen, L. Xiao, B. Tian and W. Wu, "MAFF-Net: A Deep Fusion Framework for Multimodal Recognition of LSS Targets," in *IEEE Transactions on Aerospace and Electronic Systems*.